

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical – I

Roll No.: S011	Name: Harshita Kanojia
Class: TYIT	Batch: 1
Date of Assignment: 20/06/2026	Date/Time of Submission: 20/06/2026

AIM: Environment Setup and JavaScript Basics

A. Install Node.js and Visual Studio Code, and verify installation.

OUTPUT:

```
● himanimalankar@Himanis-Macbook MERN Practical % node -v
v24.17.0
● himanimalankar@Himanis-Macbook MERN Practical % npm -v
11.13.0
❖ himanimalankar@Himanis-Macbook MERN Practical %
```

B. Create a simple Node.js file and run it using the terminal.

CODE:

```
console.log("Hello Welcome to 1st practical of MERN!!!")  
console.log("Harshita Kanojia")
```

CODE DESCRIPTION:

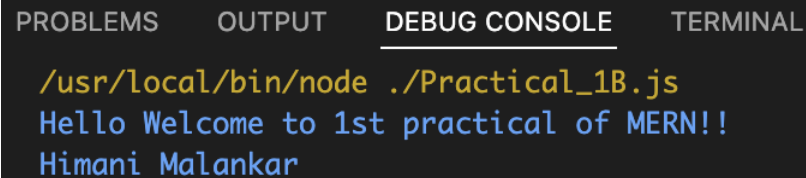
1. `console.log()` is used to display output on the console.
2. The first statement displays a welcome message.
3. The second statement displays the user's name.
4. Strings are written inside double quotes.
5. The program demonstrates basic output printing in Node.js.

FUNCTION USED:

console.log()

- Displays messages or values on the console.
- Commonly used for output and debugging.

OUTPUT:



The screenshot shows a terminal window with a dark background. At the top, there are four tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE' (which is selected and underlined), and 'TERMINAL'. Below the tabs, the terminal displays the command `/usr/local/bin/node ./Practical_1B.js` in yellow. The output consists of two lines: `Hello Welcome to 1st practical of MERN!!` and `Himani Malankar`, both in blue.

C. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

CODE:

```
console.log("Basic Calculator...")

//Assigning Values
let a = 11;
let b = 11;

console.log("Addition: " , a+b)
console.log("Substraction: " , a-b)
console.log("Multiplication: ",a*b)
console.log("Divison: ",a/b)
console.log("Modulas: ",a%b)
```

CODE DESCRIPTION:

1. The program displays a heading “**Basic Calculator...**” on the console.
2. Two variables, **a** and **b**, are assigned the value 11.
3. The program performs addition of the two numbers.
4. The program performs subtraction of the two numbers.
5. The program performs multiplication of the two numbers.
6. The program performs division of the two numbers.
7. The program calculates the modulus (remainder) of the two numbers.
8. The results of all operations are displayed on the console.

FUNCTIONS USED:

console.log()

- Displays messages and results on the console.

Variables (let)

- Used to store values that can be used in calculations.

OUTPUT:

```
[Running] node "/Users/himanimalankar/Desktop/MERN Practical/Practical_1C.js"
Basic Calculator...
Addition: 22
Substraction: 0
Multiplication: 121
Divison: 1
Modulas: 0
```

D. Create and run a simple program using variables and functions.

CODE:

```
let num1 = 15;
let num2 = 25;

function add(a , b){
    return a + b;
}

function multiply(a , b){
    return a * b;
}

console.log("1st number: ", num1);
console.log("2nd number: ",num2);

console.log("Addition = ", add(num1 , num2));
console.log("Multiplication = ", multiply(num1 , num2))
```

CODE DESCRIPTION:

1. Two variables, **num1** and **num2**, are used to store the values 15 and 25.
2. The **add()** function is used to calculate the sum of two numbers.
3. The **multiply()** function is used to calculate the product of two numbers.
4. The program displays the values of both numbers on the console.
5. The **add()** function is called to perform addition and display the result.
6. The **multiply()** function is called to perform multiplication and display the result.
7. The program demonstrates the use of **functions, variables, and output statements** in JavaScript.

FUNCTIONS USED:

add()

- Returns the sum of two numbers.

multiply()

- Returns the product of two numbers.

console.log()

- Displays messages and results on the console.

OUTPUT:

```
[Running] node "/Users/himanimalankar/Desktop/MERN Practical/Practical_1D.js"
1st number: 15
2nd number: 25
Addition = 40
Multiplication = 375
```

E. Demonstrate use of operators and control statements in JavaScript

CODE:

```
let a = 30;
let b = 20;
let day = 2;

//Comparison Operators
console.log("\nComparison Operators");
console.log("a > b :", a > b);
console.log("a < b :", a < b);
console.log("a == b :", a == b);
console.log("a != b :", a != b);

//if-else Statement
console.log("\nIf-Else Statement");
if (a > b){
    console.log("a is greater than b");
}else{
    console.log("b is greater than a or equal to a");
}

console.log("\nSwitch Statement");
switch(day){
    case 1:
        console.log("Monday");
        break;

    case 2:
        console.log("Tuesday");
        break;

    case 3:
        console.log("Wednesday");
        break;

    default:
        console.log("Invalid Day");
}
```

CODE DESCRIPTION:

1. Three variables, **a**, **b**, and **day**, are initialized with values.
2. The program demonstrates the use of **comparison operators**.
3. It compares the values of **a** and **b** and displays the results.

4. The **if-else statement** checks whether **a** is greater than **b** and displays the appropriate message.
5. The **switch statement** checks the value of **day**.
6. Based on the value of **day**, it displays the corresponding day name.
7. If the value does not match any case, the **default** statement is executed.
8. The program demonstrates **comparison operators, conditional statements, and switch-case statements** in JavaScript.

FUNCTIONS USED:

console.log()

- Displays messages and results on the console.

if-else

- Used to make decisions based on a condition.

switch

- Used to select and execute a block of code from multiple options.

OUTPUT:

```
[Running] node "/Users/himantalankar/Desktop/MERN Practical/Practical_1E.js"

Comparison Operators
a > b : true
a < b : false
a == b : false
a != b : true

If-Else Statement
a is greater than b

Switch Statement
Tuesday
```

F. Write programs using loops and template literals.

CODE:

```
let num = 20;

console.log(`Multiplication Table of ${num}`);

for (let i = 1 ; i <=10 ; i++){

    console.log(`${num} X ${i} = ${num * i}`);

}
```

CODE DESCRIPTION:

1. The variable **num** is assigned the value 20.
2. The program displays a heading for the multiplication table.
3. A **for loop** is used to repeat the statements 10 times.
4. The loop variable **i** starts from 1 and increases up to 10.
5. In each iteration, the program calculates the product of **num** and **i**.
6. The multiplication table of 20 is displayed on the console.
7. The program demonstrates the use of **variables, loops, and template literals** in JavaScript.

FUNCTIONS USED:

console.log()

- Displays messages and results on the console.

for loop

- Repeats a block of code a specified number of times.

Template Literals

- Used to insert variables and expressions into a string.

OUTPUT: